

The Idempotence Separation

Why an Admissibility Layer Cannot Carry an Accumulating Cost, and the Exact Scope of That Prohibition

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Extracted from the errata tables of "May, Does, Seen" and "The Two Bills," where the same one-line result was recorded twice as a correction. Every number is quoted verbatim from live computation performed for this paper. The result is stated here in its general form and its scope is fixed by exhaustive computation, because the two previous statements of it were both withdrawn for overreach.

Abstract

Two papers in this program independently discovered, and independently mis-stated, the same one-line theorem. Both recorded the correction in an errata table rather than in the body. This paper extracts it, states it in the form under which it cannot be attacked, and fixes its scope by exhaustive computation.

The theorem: there is no nonzero homomorphism from an idempotent semigroup into the additive reals. If x composed with x equals x , and f carries composition to addition, then $f(x) = 2f(x)$, hence $f(x) = 0$. No lattice structure is required, and stating it as a lattice result is what caused both prior withdrawals — lattices do carry valuations, and the first version of this claim denied it.

The scope is measured rather than argued. On the power set of four elements, the space of meet-homomorphisms into the reals has dimension exactly 0, while the space of modular valuations satisfying inclusion-exclusion has dimension exactly 5. The prohibition is on homomorphic additivity under meet and on nothing else.

A second register of the same result is then established, and it is the mechanism rather than the algebra. Among the 4^1 idempotent maps on four states, every second application carries a

conditional entropy of exactly zero under the pushforward measure, with maximum drift 0.000e+00. The repeat is free not by convention but because the measure that reaches the second boundary has already been collapsed by the first. The algebraic prohibition and the measured saturation are one statement.

The paper is explicit about what the theorem does not deliver. It is a separation result, not an existence result. It proves that an accumulating cost cannot live on an idempotent layer. It does not prove that a second layer exists, and it does not prove that the second layer's cost is additive — that additivity was measured, in eight of eight chains, and is carried here as MEASURED. The two-ledger architecture is FORCED only when the lemma is combined with that measurement, and the earlier framing which called the decomposition forced by the lemma alone is retracted in Section 8.

1 Where This Paper Starts

"May, Does, Seen" reported that the interface layer refused to join the ledger the preceding paper had built, and located the refusal in one line of algebra. Its errata table then recorded that the line had first been written wrongly: the claim that a lattice admits no additive valuation unless it is graded was withdrawn as false as stated, and replaced by the observation that idempotence alone suffices.

"The Two Bills" recorded the identical correction, in the identical language, in its own error table. Two papers, one result, twice logged as an erratum and never once stated as a theorem.

That is a duplication of representation rather than of content, and the program's own auditing discipline says what to do about it. The line is not an erratum. It is the reason the two bills exist. Everything downstream — why filters compose by conditioning rather than multiplication, why the admission price is paid once, why the constraint layer cannot carry a toll — is a corollary of it. This paper gives it a name and a scope.

2 The Lemma

Definition. A band is a semigroup in which every element is idempotent: x composed with x equals x , for all x . A meet-semilattice is a band that is also commutative. The set of subsets of a finite set under intersection is a meet-semilattice, and so is the set of admissible behaviours under conjunction of constraints.

Lemma (Idempotence Separation) [PROVEN]. *Let S be a band and let $f: S$ to R satisfy $f(x$ composed with $y) = f(x) + f(y)$ for all x, y in S . Then f is identically zero.*

Proof. Fix x in S . Idempotence gives x composed with $x = x$, so $f(x) = f(x \text{ composed with } x)$. Additivity gives $f(x \text{ composed with } x) = f(x) + f(x) = 2f(x)$. Therefore $f(x) = 2f(x)$, hence $f(x) = 0$. The element x was arbitrary. Square.

The proof uses one equation and one substitution. It requires no order structure, no distributivity, no grading, no finiteness, and no commutativity. This is why the general form is the safe form: every hypothesis that could be attacked has been removed, because none of them were used.

Corollary. No layer whose composition law is idempotent can carry a cost that accumulates under composition. Any attempt to define one yields the zero function.

3 The Scope, Fixed by Computation

Both previous statements of this result overreached in the same direction: they denied more than the lemma denies. Lattices carry valuations in abundance. Cardinality is one. Any measure is one. The distinction that must be held is between two different additivity conditions, and the cleanest way to hold it is to count the solutions of each.

On the power set of a four-element set — 16 nodes — two linear systems were solved over all 256 ordered pairs of subsets.

Condition imposed on f	Constraints	Rank	Solution dim
$f(A \wedge B) = f(A) + f(B)$	256	16	0
$f(A) + f(B) = f(A \wedge B) + f(A \vee B)$	256	11	5

The first row is the lemma, confirmed exhaustively: the only meet-homomorphism into the additive reals is the zero function. The second row is the modular valuation condition, and its solution space has dimension five — one degree of freedom for the empty set and one for each of the four singletons, which is exactly the family containing cardinality and every measure on four atoms.

The prohibition is therefore narrow and exact. It forbids homomorphic additivity under meet. It does not forbid valuations, it does not forbid monotone functions, and it does not forbid the layer from being measured at all. An admissibility layer can be assigned a size; what it cannot be assigned is a bill that grows when the same constraint is applied twice.

4 The Second Register: Why the Repeat Is Free

The lemma says an accumulating cost on an idempotent layer must be zero. It does not by itself say what happens physically when an idempotent operation is applied twice. That is a separate question with a separate answer, and the two answers turn out to be the same statement read in two registers.

Take the crossing price of the predecessor paper, $X(f) = H(X|Y)$, with the intermediate carrying the pushforward measure. Among all 256 maps on four states, 41 are idempotent. For each, the toll of the first application was computed under the uniform source, and the toll of the second was computed under the measure the first application produced.

Measurement	Result
Idempotent maps on four states	41 of 256
Maximum $ X(f \circ f) - (X(f) + X_{\text{push}}(f)) $	0.000e+00
Idempotents with nonzero second-pass toll	0 of 41
Chain rule drift, 3,000 random pairs	0.000e+00
Naive same-measure doubling error	max 2.0000 bits, mean 0.9821 bits

Every idempotent operation pays exactly once, and the second pass is free to machine exactness in all 41 cases. The mechanism is visible in the third row: the first application has already collapsed the source measure onto its image, and the second application sees a measure on which it acts injectively. There is no fibre left to charge for.

This is the same phenomenon the predecessor measured as $P(F_2 | F_1) = 1.0000$ and reported as filter saturation. It is now derived rather than observed: an idempotent map is, by definition, the identity on its own image, and the pushforward measure is supported on that image.

The two registers. Algebraically, an additive homomorphism on a band must vanish. Information-theoretically, the second application of an idempotent map has zero conditional entropy under the measure the first one produced. The first is a statement about what cost functions can exist; the second is a statement about why the cost function that does exist returns zero. They are not two results. They are one result seen from the constraint side and from the execution side, which is what the two-ladder method was built to detect.

5 What the Lemma Does Not Prove

This section exists because the previous two statements of the lemma were withdrawn for claiming too much, and the natural third overreach is available and tempting.

The tempting statement is that the two-bill decomposition is forced by the lemma. It is not. The lemma is a separation result. It establishes:

That an accumulating cost cannot live on an idempotent layer. **PROVEN**, by Section 2, confirmed by Section 3.

It does not establish either of the following, both of which are required before the architecture can be called forced:

That a second, non-idempotent layer exists at all. **Supplied by C1** — transitions occur, and the monoid of maps on four states has 41 idempotent elements out of 256, so it is not a band. Composition of general maps accumulates because most maps are not idempotent, which is a fact about the monoid and not a consequence of the lemma.

That the cost on the second layer is additive. **MEASURED** — the chain rule was verified in eight of eight chains to nine decimals in the predecessor and at drift 0.000e+00 over 3,000 pairs here. Additivity of the crossing price is an empirical property of conditional entropy under pushforward, not something the lemma delivers.

The correct compound statement. Given a layer whose composition is idempotent, and given a second layer on which cost is measured to accumulate, the accounting must split. The split is FORCED by the conjunction. It is not forced by the lemma alone, and any sentence asserting that it is should be read as this paper's own error, corrected here before it reached a body section.

6 The Architecture That Follows

With the labels attached correctly, the architecture reads as follows.

	Idempotent layer	Accumulating layer
Composition	Meet, conjunction, intersection	Sequential composition of maps
Algebraic type	Band (idempotent semigroup)	Monoid, not a band
Cost behaviour	Paid once; repeat is free	Paid per crossing; telescopes

	Idempotent layer	Accumulating layer
Admissible accounting	Valuation, not homomorphism	Additive homomorphism
Instance: prices	Admission price $A(I)$	Crossing price $X(I)$
Instance: algebras	Distributive constraint lattice	Execution monoid
Instance: interfaces	Filter	Meter
Instance: channels	Value channel, committed residue	Shape channel, propagation
Governing constraint	History preservation	Continuation

The final two rows are an identification across lineages and are carried at a weaker status than the rest. The shape and value channels were compiled in the separate primitive-set lineage from the observable event primitive, and the correspondence between the value channel and the idempotent layer is structural: both commit once, both preserve rather than accumulate, both are governed by history rather than by continuation. Whether the identification is exact — whether the value channel is a band under its own composition — has not been computed, and the row is labelled IDENTIFIED rather than PROVEN in the ledger.

7 A Falsifiable Prediction

The prediction named in the predecessor as checkable and left undone is sharpened here to a signature. If a real verification or constraint mechanism implements an idempotent composition law, then by the lemma it cannot charge an accumulating cost, and by Section 4 the second application must be free. The observable consequences are:

Re-executing a constraint operation on an already-constrained input costs nothing beyond dispatch — no additional narrowing, no additional cycles attributable to narrowing.

Constraint operations are shrinking: the output admissible set is contained in the input admissible set, and applying the operation to its own output is the identity.

Any instruction that charges twice for the same constraint is not implementing a band, and the layer it belongs to is an execution layer that has been mislabelled.

This is checkable against instruction-set documentation and against static analyzer behaviour, and it remains undone. It is the most immediately actionable item this paper produces.

8 Retractions

Withdrawn claim	Replaced by
A lattice admits no additive valuation unless it is graded (first statement, "May, Does, Seen")	False as stated; valuation space on the four-element power set has dimension 5. The correct object is the meet-homomorphism, whose solution space has dimension 0
The same claim, restated in "The Two Bills" errata	Superseded by the band form of Section 2, which uses neither lattice structure nor grading
The two-bill decomposition is forced by idempotence	Separation only. Existence of the second layer comes from C_1 ; additivity of its cost is MEASURED. The architecture is forced by the conjunction, not by the lemma
The lemma is a lattice result	It is a band result. Commutativity and order were never used in the proof

9 What Remains Open

Whether the value channel of the primitive-set lineage is a band under its own composition law, which would upgrade the last two rows of the Section 6 table from IDENTIFIED to PROVEN and would connect the two lineages at the accounting layer rather than only by analogy.

Whether the admission price behaves under a genuine lattice join. The predecessor left this open and predicted asymmetry with the conjunctive case; the lemma does not settle it, because join is also idempotent and the lemma therefore forbids an accumulating cost on that side too, without saying what the non-accumulating cost is.

Whether any physical constraint mechanism exhibits the Section 7 signature. Checkable, not checked.

Whether the separation extends to layers that are idempotent only up to equivalence — near-bands, where x composed with x is equivalent to but not equal to x . The proof breaks immediately, since $f(x) = 2f(x)$ requires equality, and the physically interesting cases may all be of this kind.

And permanently: the derivation is never completed. A finished self-description is a map identical to its territory. The prohibition is carried forward from the predecessor papers with one correction noted here, since a committed measurement is a frozen projection whose complementary channel continues, and the forbidden object is the one that freezes both. That repair belongs to the ledgers it appears in and is not executed in this paper.

Status Ledger

Claim	Status	Basis (live numbers)
No nonzero homomorphism from a band to $(R, +)$	PROVEN	One line; no hypotheses beyond idempotence
Meet-homomorphism space on 2^4 is trivial	PROVEN	256 constraints, rank 16, solution dim 0
Modular valuations survive; space is 5-dimensional	PROVEN	256 constraints, rank 11, solution dim 5
Second application of an idempotent map is free	PROVEN	41 of 41; drift 0.000e+00
Chain rule under pushforward measure	PROVEN	3,000 pairs; drift 0.000e+00
Naive same-measure accounting overcharges	MEASURED	max 2.0000 bits, mean 0.9821 bits
Execution monoid is not a band	PROVEN	41 idempotents of 256 maps on 4 states
Additivity of the crossing price	MEASURED	Not derived; 8 of 8 chains in predecessor
Two-ledger split forced by the lemma alone	RETRACTED	Separation only; see Section 5
Value channel is the idempotent layer	IDENTIFIED	Structural correspondence across lineages
Idempotent instruction signature in real hardware	PREDICTED	Falsifiable; not checked

Claim	Status	Basis (live numbers)
Admission price under lattice join	OPEN	Lemma forbids accumulation; form unknown
Near-bands (idempotent up to equivalence)	OPEN	Proof requires strict equality
Derivation ever completed	FORBIDDEN	Complete self-description freezes both channels

Appendix: Live Output, Quoted Verbatim

Scope determination — dimension of each solution space on the power set of four elements.

lattice: power set of 4 elements -> 16 nodes

(1) meet-homomorphism $f(A \wedge B) = f(A) + f(B)$: constraints 256 rank 16 solution dim 0

(2) modular valuation $f(A) + f(B) = f(A \wedge B) + f(A \vee B)$: constraints 256 rank 11 solution dim 5

(3) monoid of all maps on 4 states: $|M| = 256$ idempotent elements = 41 (band would require 256)

Second register — the cost of the repeat, under the pushforward measure.

idempotents tested : 41

$\max |X(f \circ f) - (X(f) + X_{\text{push}}(f))|$: 0.000e+00

idempotents with nonzero second-pass toll: 0

chain rule $X(g \circ f) = X(f) + X_{\text{push}}(g)$, 3000 pairs, max drift: 0.000e+00

naive doubling error on idempotents: max 2.0000 mean 0.9821

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